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DYNAMIC PARAMETER ADAPTATION IN PARETO-OPTIMAL PARTICLE SWARM OPTIMIZATION FOR EFFICIENT UAV PATH PLANNING

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Abstract. Efficacious path planning is crucial for the autonomous operations of Unmanned Aerial Vehicles (UAVs) in complex real-world scenarios. The existing path-planning approaches often struggle to balance multiple objectives under dynamic environmental conditions, resulting in suboptimal mission performance. To overcome these limitations, this study proposes an enhanced UAV path-planning algorithm that integrates multiple strategies within a unified framework. The path-planning task is formulated as a multi-objective optimization problem and solved using the Pareto Optimal Particle Swarm Optimization (POPSO) algorithm. The proposed method incorporates hybrid random–deterministic initialization to improve solution diversity, a multi-objective cost function with dynamic constraints to adapt to real-time environmental changes, dynamic updating of PSO parameters for adaptive optimization, and a hybrid Pareto-entropy selection mechanism to balance quality and diversity. Experimental results show that the proposed algorithm achieves enhanced performance in generating a well-distributed set of Pareto-optimal solutions. A comparison of simulations with other state-of-the-art multi-objective methods indicates that the proposed approach improves path efficiency, safety, and adaptability, making it well-suited for complex UAV mission execution.

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1. Introduction

Unmanned Aerial Vehicles (UAVs) have become indispensable in various fields, including environmental monitoring, disaster response, surveillance, and many others. The success of UAV missions relies on effective path planning, which must balance multiple objectives such as minimizing path length, ensuring obstacle avoidance, and optimizing energy efficiency. It is noteworthy that UAV path planning remains a challenging problem due to dynamic environmental factors, including moving obstacles and varying weather conditions [1,2]. In applications such as aerial photography, mapping, and search-and-rescue operations, an optimal trajectory is often defined by minimizing travel distance to reduce energy consumption and flight time [3, 4]. For surveillance missions, maximizing coverage while minimizing flight time is crucial. In such multi-objective scenarios, identifying Pareto-optimal solutions is essential for achieving a balanced trade-off among conflicting objectives. Additionally, the trajectory must be both feasible and safe, satisfying operational constraints [5].

Keywords. UAV path planning, Pareto-optimal PSO, multi-objective cost function, Pareto-entropy, dynamic constraints.

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Several methodologies for UAV path planning have been explored, ranging from traditional approaches such as roadmap-based methods, spatial partitioning techniques, and artificial potential fields to advanced nature-inspired optimization techniques. Among these, bio-inspired algorithms have gained significant attention due to their ability to efficiently explore high-dimensional search spaces and avoid local optima. Metaheuristic techniques, including Cuckoo Search [6], Genetic Algorithm (GA) [7], Differential Evolution (DE) [8], Artificial Bee Colony (ABC) [9], Ant Colony Optimization (ACO) [10], and Particle Swarm Optimization (PSO) [11], have been widely applied to UAV trajectory optimization. However, classical methods such as GA and PSO often face challenges in real-time navigation involving conflicting objectives, particularly in balancing exploration and exploitation, which can lead to suboptimal path selection in dynamic environments [12,13]. To overcome these issues, hybrid approaches integrating PSO with Reinforcement Learning (RL) [14], Deep Learning (DL) [15], and Deep Reinforcement Learning (DRL) [16] models have also been investigated to enhance adaptability. Among these, PSO remains one of the most widely utilized techniques, with several adaptive and multi-objective variants developed to improve convergence speed and path efficiency [17,18].

Multi-objective evolutionary algorithms (MOEAs) have been introduced to better address such challenges by identifying Pareto-optimal solutions. Yet, MOEAs are often limited by slow convergence and reduced adaptability in real-time environments [19]. Within this class, Multi-Objective Particle Swarm Optimization (MOPSO) [20–23] has emerged as a popular method for solving UAV path planning with conflicting objectives and complex real-world constraints. Building on this foundation, the proposed work enhances the MOPSO framework.

In UAV path planning, prior studies have employed multi-objective cost functions that consider path length, energy efficiency, safety, obstacle avoidance, and weather conditions. For instance, MOPSO has been applied to minimize path length and safety risks [24–26], algorithms have been designed to generate minimum-length collision-free paths in 3D urban environments [27–29], and models have been proposed to jointly minimize path length, energy consumption, and risk [30,31]. However, integrating all these factors into a single dynamic cost function remains relatively uncommon. To address this gap, this work introduces a new objective function formulation along with an efficient methodology for solving complex UAV path-planning problems.

The proposed POPSO framework for UAV path planning simultaneously optimizes conflicting objectives: path length, energy consumption, safety, obstacle avoidance, and weather adaptation. To improve performance, the methodology integrates four key strategies. A hybrid initialization scheme combining grid-based and random placement ensures well-distributed initial solutions and improves exploration in complex environments [32,33]. A dynamic multi-objective cost function adapts to mission-specific requirements and environmental variations, enabling robust trajectory generation [34,35]. Adaptive parameter control updates the inertia weight and learning coefficients during optimization to accelerate convergence and mitigate stagnation [36]. Finally, a Pareto-entropy selection mechanism combines dominance-based classification with entropy-guided diversity preservation to maintain a well-distributed high-quality set of solutions. Within this Pareto-based framework, each objective is tracked independently, and non-dominated solutions are maintained in an external archive, from which entropy-based selection is used to efficiently approximate the Pareto front.

The major contributions of this work are as follows.

(i) UAV path planning is formulated as a multi-objective optimization problem with five objectives: path length, energy efficiency, safety, obstacle avoidance, and weather adaptation, addressed through a dynamic cost function that captures their conflicting nature and adapts to real-time environmental constraints.
(ii) An enhanced POPSO algorithm is proposed that integrates hybrid random-deterministic initialization, dynamic parameter adaptation, and entropy-guided selection within a Pareto-based PSO, ensuring high-quality solutions and robust optimization.
(iii) POPSO is benchmarked against NSGA-II, MODE, MOEA/D, classical PSO, GA, and others in diverse terrains and threat distributions, demonstrating significant performance gains.

The rest of the paper is organized as follows. Section 2 presents the related work. The proposed methodology for UAV route planning is explained in Section 3. The simulation results are shown in Section 4. In Section 5,

the effectiveness and benefits of the POPSO-based strategy are discussed along with a comparison study of the suggested technique with existing algorithms of a similar kind. The paper is finally concluded in Section 6 with an overview of the major discoveries and contributions.

2. Related work

Multi-objective particle swarm optimization (MOPSO) has been widely employed in UAV path planning, where multiple conflicting objectives such as path length, safety, smoothness, and energy consumption must be balanced. Duong *et al.* proposed the Navigation Variable-based Multi-objective Particle Swarm Optimization (NMOPSO), which integrates kinematic constraints and adaptive mutation to enhance swarm diversity and path feasibility in UAV trajectory planning [37]. Coello *et al.* [38] pioneered the Multi-Objective Particle Swarm Optimization (MOPSO) approach by introducing Pareto dominance and an external archive to maintain solution diversity. Their method laid the foundation for applying MOPSO to complex tasks such as UAV path planning, where multiple conflicting objectives must be optimized simultaneously. More recently, Quamar *et al.* combined reinforcement learning with MOPSO to adapt UAV routing to dynamic environmental changes, showing that hybridized approaches can enhance real-time decision making [39]. These works establish the value of MOPSO in aerial robotics but often remain limited to static or semi-dynamic conditions, without fully exploiting adaptive parameter control or entropy-based diversity mechanisms.

Beyond UAV applications, MOPSO has been successfully used across a range of optimization problems. In feature selection, Hebie *et al.* developed a fuzzy-cost MOPSO framework that achieved superior subset quality on high-dimensional datasets [40], while Li and Chen introduced a hybrid MOPSO variant for classification tasks with strong accuracy improvements [41]. In cloud and edge computing, Liu *et al.* applied multi-objective Cauchy PSO for energy-aware virtual machine placement [42], and Zhong and Wang proposed a parallel MOPSO for container-based microservice scheduling to handle large-scale workloads efficiently [43]. Wireless sensor networks (WSN) have also benefited from MOPSO: Tian *et al.* optimized dual cluster-head selection using Pareto-based MOPSO for improved network lifetime [44], Tong *et al.* enhanced coverage and node minimization using a hybrid PSO approach [45], and Zhang *et al.* applied improved PSO to design energy-efficient adaptive routing strategies in heterogeneous WSNs [46]. These applications highlight the versatility of MOPSO in addressing trade-offs across diverse domains.

Despite these advances, most existing studies have addressed static or slowly varying optimization problems, where objectives and constraints remain fixed during the optimization process. UAV path planning, in contrast, requires handling dynamic, non-stationary conditions such as moving obstacles, uncertain weather, and changing mission objectives. This motivates the need for enhanced algorithms that adapt parameters online and maintain diversity in rapidly changing search spaces. Our work contributes to this gap by integrating dynamic parameter adaptation with an entropy-guided Pareto archiving mechanism within a unified MOPSO framework.

3. Methodology

Figure 1 presents the proposed POPSO framework for UAV path planning, which consists of three main modules. First, terrain data from satellite imagery or digital elevation models (DEMs) is processed to extract obstacles and spatial features. Second, POPSO performs dynamic multi-objective optimization, evaluating paths by length, energy, safety, obstacles, and weather. Initialization combines grid-based and random placement for balanced exploration, while adaptive parameters guide swarm updates. Pareto dominance with entropy-based selection maintains diversity and robustness. Finally, the optimized UAV path is visualized on the terrain map, highlighting obstacle avoidance and efficiency.

3.1. Cost function formulation

This work proposes a multi-objective cost function with dynamic constraints that adapts to environmental changes such as moving obstacles and varying weather conditions to improve UAV path planning. Adjusting the

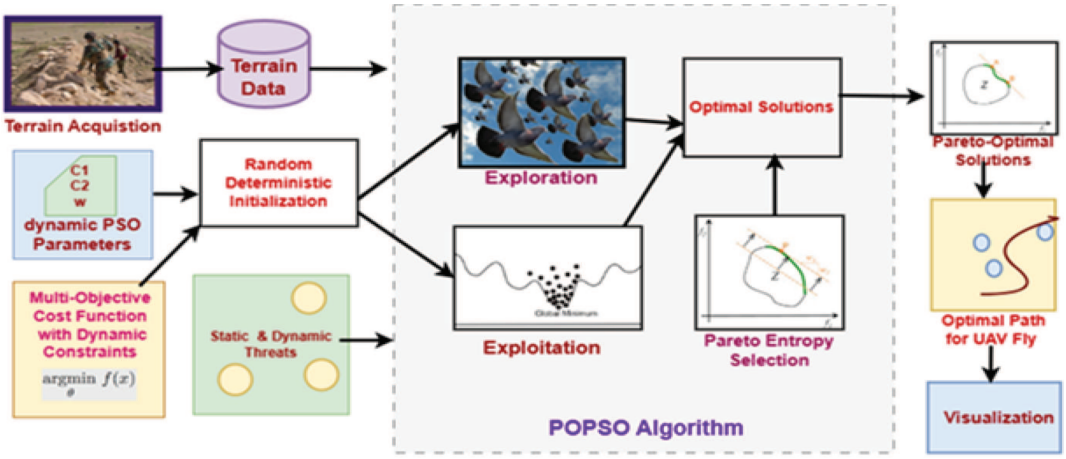


FIGURE 1. The proposed POPSO architecture.

evaluation criteria in real time [47], the approach improves robustness and flexibility. The function simultaneously optimizes multiple conflicting objectives, such as path length, energy efficiency, safety, obstacle avoidance, and weather adaptation, while dynamically adjusting to real-time environmental variations. These objectives, which capture the UAV's kinematics and help facilitate efficient maneuverability, are formulated as follows.

3.1.1. Basic cost components

Let the cost function $F(\mathbf{x}_i)$ for a particle $\mathbf{x}_i$ be defined as a weighted sum of three main objectives:

- **Path Length** ($L(\mathbf{x}_i)$): The total distance the UAV must travel.
- **Energy Efficiency** ($E(\mathbf{x}_i)$): The energy consumption required for the UAV to complete its path.
- **Safety** ($S(\mathbf{x}_i)$): A measure of the risk of collisions or flying through hazardous areas.

The cost function is given by:

$$F(\mathbf{x}_i) = w_1 \cdot L(\mathbf{x}_i) + w_2 \cdot E(\mathbf{x}_i) + w_3 \cdot \frac{1}{S(\mathbf{x}_i)} \tag{1}$$

where w_1, w_2, and w_3 are weights assigned to each objective, reflecting their relative importance.

3.1.2. Incorporation of dynamic constraints

To account for dynamic changes in the environment, additional objectives are introduced in the cost function that adapts to real-time conditions:

- **Obstacle Avoidance Penalty** ($O(\mathbf{x}_i, t)$): A penalty term that increases when the path intersects with moving obstacles detected at time t.

$$O(\mathbf{x}_i, t) = \sum_{j=1}^{M} \exp\left(-\frac{d_{ij}^2(t)}{\sigma^2}\right) \tag{2}$$

where $d_{ij}(t)$ is the distance between the UAV's path and the j-th obstacle at time t, and σ is a scaling factor controlling the sensitivity to obstacles.

– **Weather Condition Factor** ($W(\mathbf{x}_i, t)$): A factor that adjusts based on real-time weather data, such as wind speed and direction, that may affect the UAV's energy consumption and safety.

$$W(\mathbf{x}_i, t) = \gamma \cdot \left(1 + \frac{v_{\text{wind}}(t)}{v_{\max}}\right) \tag{3}$$

where $v_{\text{wind}}(t)$ is the wind speed at time t, $v_{\max}$ is a reference maximum wind speed, and γ adjusts the impact of weather conditions.

3.1.3. Objective vector definition

Each candidate UAV path (particle) $\mathbf{x}_i$ is evaluated as a multi-objective vector:

$$\mathbf{f}(\mathbf{x}_i, t) = \left[L(\mathbf{x}_i), E(\mathbf{x}_i, t), \tilde{S}(\mathbf{x}_i, t), O(\mathbf{x}_i, t), W(\mathbf{x}_i, t)\right]^T, \tag{4}$$

where:

– $L(\mathbf{x}_i)$ – path length (distance travelled by UAV),
– $E(\mathbf{x}_i, t)$ – total energy consumption,
– $\tilde{S}(\mathbf{x}_i, t)$ – inverse safety term,
– $O(\mathbf{x}_i, t)$ – obstacle avoidance penalty,
– $W(\mathbf{x}_i, t)$ – weather adaptation factor,
– t – iteration or mission time index.

This vector form enables Pareto-based dominance comparison and entropy-guided diversity maintenance (refer to Algorithm 3).

3.1.4. Objective normalization

Since the objectives differ in scale and units, they are normalized to the range $[0, 1]$ for balanced comparison:

$$\hat{f}_j(\mathbf{x}_i, t) = \frac{f_j(\mathbf{x}_i, t) - f_j^{\min}}{f_j^{\max} - f_j^{\min} + \epsilon}, \tag{5}$$

where $f_j^{\min}$ and $f_j^{\max}$ denote the lower and upper reference bounds (obtained from preliminary runs or theoretical limits), and $\epsilon = 10^{-8}$ avoids division by zero.

The safety objective is redefined to avoid singularities:

$$\tilde{S}(\mathbf{x}_i, t) = \frac{1}{S(\mathbf{x}_i, t) + \epsilon_S}, \tag{6}$$

where ϵ_S is a small constant that prevents numerical instability.

3.1.5. Dynamic weighted cost function

The cost function is formulated as a weighted sum [48–50] of multiple objectives based on the scalarization approach in multi-objective optimization problems. For scalar-based position updates and final trajectory selection, an auxiliary weighted-sum fitness function is defined using normalized objectives:

$$F_w(\mathbf{x}_i, t) = \sum_{j=1}^{5} \alpha_j \hat{f}_j(\mathbf{x}_i, t), \quad \sum_{j=1}^{5} \alpha_j = 1, \quad \alpha_j \geq 0. \tag{7}$$

Here, α_j represents the relative importance of each objective, typically determined empirically based on mission priorities. This weighted sum serves two auxiliary purposes:

(i) provides scalar fitness values for particle updates during optimization, and
(ii) facilitates selection of a representative solution from the Pareto archive for mission deployment.

However, the overall optimization remains fundamentally multi-objective, where Pareto dominance and entropy-based selection govern the external archive.

3.1.6. Constraints

The optimization problem is subject to UAV operational and environmental limits:

$$\text{Path constraints: } \mathbf{x}_i(t) \in \mathcal{X}_{\text{safe}}, \quad \forall t \in [0, T], \tag{8}$$

$$\text{Energy constraint: } E_{\text{total}}(\mathbf{x}_i, t) \leq E_{\max}, \tag{9}$$

$$\text{Obstacle constraint: } d_{ij}(t) \geq d_{\min}, \quad \forall j = 1, \ldots, M, \tag{10}$$

$$\text{Boundary limits: } X_{\min} \leq x_i(t) \leq X_{\max}, \;\; Y_{\min} \leq y_i(t) \leq Y_{\max}. \tag{11}$$

where, $\mathbf{x}_i(t)$ denotes the position of the UAV at time t, $\mathcal{X}_{\text{safe}}$ the feasible flight region, T is the planning horizon and $E_{\max}$ is the maximum available energy. Constraint violations are penalized as

$$P_{\text{viol}}(\mathbf{x}_i) = \kappa \sum_{k,j} \max(0, d_{\min} - d_{ij,k}), \tag{12}$$

and the penalized objective is $F'_w = F_w + P_{\text{viol}}$, where κ is a large penalty coefficient ensuring feasibility.

3.1.7. Multi-objective optimization problem

The final optimization problem can thus be expressed as:

$$\min_{\mathbf{x}_i} \quad \mathbf{f}(\mathbf{x}_i, t), \tag{13}$$

$$\text{subject to:} \quad \text{constraints (8)}-\text{(11)}. \tag{14}$$

During optimization, the Pareto engine uses $\mathbf{f}(\mathbf{x}_i, t)$ for dominance comparison and diversity maintenance, while $F_w(\mathbf{x}_i, t)$ guides velocity updates and decision making. The cost function framework ensures that POPSO efficiently balances multiple competing criteria in dynamic environments.

3.2. The proposed (POPSO) algorithm

The Pareto-Optimal Particle Swarm Optimization (POPSO) algorithm enhances the standard PSO by incorporating multi-objective optimization capabilities. It maintains a diverse set of non-dominated solutions in an external archive and integrates hybrid initialization, adaptive parameter updates, and entropy-guided selection to improve exploration and convergence. While this multi-objective engine runs, a weighted sum approach provides scalar fitness values to guide individual particles and serves as a decision criterion when a single representative solution must be chosen. This hybrid methodology effectively combines broad multi-objective search with targeted single-objective selection. The overall process is summarized below and formally presented in Algorithm 1.

The proposed POPSO algorithm is structured into five core components:

(i) **Hybrid Initialization (Sect. 3.3):** A hybrid random-deterministic scheme is used to initialize the positions and velocities of particles within the feasible search space. This balances exploration through randomness with coverage from structured initialization, ensuring a well-distributed starting swarm.

(ii) **Multi-Objective Evaluation (Sect. 3.1):** Each particle is evaluated using a dynamic cost function that simultaneously accounts for multiple objectives, including path length, energy efficiency, safety, obstacle avoidance, and weather adaptation. This formulation captures the conflicting nature of UAV mission requirements under real-world conditions.

(iii) **Adaptive PSO Update (Sect. 3.4):** Particle velocities and positions are updated using adaptive PSO parameters. These parameters dynamically adjust according to swarm behavior and iteration count, enhancing convergence speed while preventing premature stagnation. The parameter sensitivity analysis is discussed in Section 4.1.

Algorithm 1. Pareto-Optimal Particle Swarm Optimization (POPSO).

Initialization:
Initialize swarm using Hybrid Random–Deterministic Initialization (Alg. 3) with position $\mathbf{x}_i$ and velocity $\mathbf{v}_i$.
Initialize an empty external archive A.
Define the dynamic multi-objective cost function $F(\mathbf{x}_i, t)$ from Eq. (7).
for each particle i **do**
  Evaluate $F(\mathbf{x}_i, t)$.
  if particle is non-dominated w.r.t. A **then**
    Insert particle into A.
  end if
end for
for each iteration t **do**
  Update PSO parameters $\omega(t), c_1(t), c_2(t)$ using Eqs. (15)–(17).
  for each particle i **do**
    Update velocity:

$$\mathbf{v}_i(t+1) = \omega(t)\mathbf{v}_i(t) + c_1(t)r_1(\mathbf{p}_i - \mathbf{x}_i(t)) + c_2(t)r_2(\mathbf{g} - \mathbf{x}_i(t)) \qquad \text{(Eq. 18)}$$

    Update position:

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \qquad \text{(Eq. 19)}$$

    Apply boundary and threat-avoidance constraints (Eqs. (8)–(11)).
  end for
  Re-evaluate each particle using Eq. (7).
  Update archive A:
  - Add non-dominated solutions.
  - Remove dominated solutions.
  - Apply Pareto–Entropy Selection (Alg. 3) to preserve diversity.
end for
Terminate if stopping criteria are met (max iterations or convergence).
return External archive A containing Pareto-optimal UAV paths. =0

(iv) **Archive Maintenance with Pareto-Entropy Selection (Sect. 3.5):** Non-dominated solutions are preserved in an external archive, while dominated ones are discarded. A Pareto entropy – based selection mechanism is applied to maintain diversity in the archive, ensuring a wide distribution of Pareto-optimal UAV trajectories.

(v) **Termination and Pareto-Optimal Path Extraction:** The evaluation-update-archive process is repeated until the stopping criterion (maximum iterations or convergence) is satisfied. The final archive provides a set of Pareto-optimal paths, offering multiple trajectory options such as shortest, safest, or most energy-efficient routes.

3.3. Hybrid random-deterministic initialization

This research introduces a Hybrid Random-Deterministic Initialization (RDI) strategy designed to enhance both diversity and efficiency in particle initialization. The deterministic component employs a grid-based structure to uniformly cover the search space, ensuring that particles are distributed in a manner that systematically avoids no-fly zones while prioritizing regions with fewer obstacles. The random component places the remaining

particles randomly within the search space, adding variability and preventing the algorithm from becoming biased toward the grid structure. This randomness boosts the algorithm's exploratory ability, helping to avoid premature convergence on local optima.

Algorithm 2. Hybrid Random-Deterministic Initialization.

Require: S (total particles), G (deterministic particles), $R = S - G$ (random particles), $X_{\min}, X_{\max}, Y_{\min}, Y_{\max}$ (boundaries), N_x, N_y (grid divisions)
Ensure: Set of initialized particles $\{(x_i, y_i)\}_{i=1}^{S}$

Step 1: Input Parameters
$R \leftarrow S - G$
Step 2: Grid-based Deterministic Initialization
Calculate Grid Intervals
$\Delta x \leftarrow \frac{X_{\max} - X_{\min}}{N_x - 1}$
$\Delta y \leftarrow \frac{Y_{\max} - Y_{\min}}{N_y - 1}$
Place Particles on Grid
Initialize particle count $g \leftarrow 0$
for $i = 0$ to $N_x - 1$ **do**
  for $j = 0$ to $N_y - 1$ **do**
    if $g < G$ **then**
      $x_i \leftarrow X_{\min} + i \cdot \Delta x$
      $y_j \leftarrow Y_{\min} + j \cdot \Delta y$
      Initialize particle at (x_i, y_j)
      $g \leftarrow g + 1$
    else
      Break
    end if
  end for
end for
Step 3: Random Initialization
for $k = 1$ to R **do**
  $x_k \sim \text{Uniform}(X_{\min}, X_{\max})$
  $y_k \sim \text{Uniform}(Y_{\min}, Y_{\max})$
  Initialize particle at (x_k, y_k)
end for
Step 4: Combine Initialized Particles
Merge grid-based and randomly initialized particles into a single set
Step 5: Output Initialized Particles
Return set of initialized particles $\{(x_i, y_i)\}_{i=1}^{S}$ =0

By combining structured placement with random sampling, the proposed approach balances coverage and exploration, ensuring that critical regions of the search space are not overlooked. In UAV path planning, this provides a robust starting point for PSO, particularly in environments with varying obstacle densities or complex terrain. The Random-Deterministic Initialization process is summarized in Algorithm 2.

3.4. Dynamic update of PSO parameters

Incorporating dynamic updates for the PSO parameters can enhance the algorithm's performance by adapting to the search space over time. The parameters that can be dynamically updated include the inertia weight (ω), cognitive coefficient (c_1), and social coefficient (c_2).

3.4.1. Dynamic inertia weight (ω)

The inertia weight balances exploration (global search) and exploitation (local search). It can be updated based on the iteration count:

$$\omega(t) = \omega_{\max} - \left(\frac{\omega_{\max} - \omega_{\min}}{\text{max_iter}} \right) \cdot t \tag{15}$$

where:

- $\omega_{\max}$ and $\omega_{\min}$ are the maximum and minimum values of inertia weight,
- t is the current iteration,
- max_iter is the maximum number of iterations.

3.4.2. Dynamic cognitive and social coefficients (c_1 and c_2)

The cognitive (c_1) and social (c_2) coefficients can also be dynamically updated to adapt the focus between personal learning and group learning:

$$c_1(t) = c_{1_{\text{start}}} + \left(\frac{c_{1_{\text{end}}} - c_{1_{\text{start}}}}{\text{max_iter}} \right) \cdot t \tag{16}$$

$$c_2(t) = c_{2_{\text{start}}} + \left(\frac{c_{2_{\text{end}}} - c_{2_{\text{start}}}}{\text{max_iter}} \right) \cdot t \tag{17}$$

where:

- $c_{1_{\text{start}}}$ and $c_{1_{\text{end}}}$ are the starting and ending values for the cognitive coefficient,
- $c_{2_{\text{start}}}$ and $c_{2_{\text{end}}}$ are the starting and ending values for the social coefficient.

3.4.3. Adaptive strategy based on performance metrics

Parameters can also be updated based on the performance of the swarm. For instance:

- Increase ω to boost exploration when the global best solution does not improve for several iterations.
- Adjust c_1 to enhance personal best exploration or c_2 to strengthen the influence of the global best. Now, the modified velocity and position update equations incorporating dynamic parameters are,

$$\mathbf{v}_i(t+1) = \omega(t)\mathbf{v}_i(t) + c_1(t)r_1\left(\mathbf{p}_i - \mathbf{x}_i(t)\right) + c_2(t)r_2\left(\mathbf{g} - \mathbf{x}_i(t)\right) \tag{18}$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \tag{19}$$

where:

- $\mathbf{v}_i(t)$ is the velocity of particle i at time t,
- $\mathbf{p}_i$ is the personal best position of particle i,
- $\mathbf{g}$ is the global best position among all particles,
- r_1 and r_2 are random values uniformly distributed in $[0, 1]$,
- $\mathbf{x}_i(t)$ is the position of particle i at time t.

3.5. Hybrid Pareto-Entropy Selection mechanism

Algorithm 3. Pareto Entropy Selection.

Require: Set of solutions P, desired number of solutions N, weight λ
Ensure: Selected subset of solutions
{Pareto dominance sorting}
$ParetoFronts \leftarrow NonDominatedSorting(P)$
{Entropy Calculation}
for each $x_i \in P$ **do**
  $distances \leftarrow [EuclideanDistance(x_i, x_j) \mid x_j \in P, x_j \neq x_i]$
  $probabilities \leftarrow [\frac{d}{\sum distances} \mid d \in distances]$
  $entropy \leftarrow -\sum p \log(p)$ for $p \in probabilities$
  $Entropies.append(entropy)$
end for
{Hybrid Selection}
$Scores \leftarrow []$
for each $x_i \in P$ **do**
  $rank \leftarrow GetParetoRank(x_i, ParetoFronts)$
  $score \leftarrow \frac{1}{(rank+1)} + \lambda \cdot Entropies[i]$
  $Scores.append((x_i, score))$
end for
{Selection}
$Scores.sort(key = \lambda x : x[1], reverse = True)$
$SelectedSolutions \leftarrow [x_i \mid x_i, _ \in Scores[: N]]$
{Output}
return $SelectedSolutions$ =0

This research proposed a Pareto-Entropy Selection [51] of final solutions that combine Pareto dominance with an entropy-based measure to select solutions. It ensures a balance between exploration and exploitation by not only considering non-dominated solutions but also maximizing the diversity of selected solutions. The Pareto Dominance Sorting is done using Non-Dominated Sorting, where P is the population of solutions. Entropies are calculated using Euclidean distances between solutions and probabilities derived from these distances. The hybrid score is calculated by combining the Pareto rank and the entropy for each solution. The solutions are sorted according to their score, and the top N solutions are selected. The selected solutions are returned. This mechanism ensures that the optimization process not only converges to high-quality solutions but also maintains a diverse set of solutions, which is crucial for the proposed multi-objective UAV path planning problem. The calculations of Pareto-entropy are given in Algorithm 3.

4. Results and discussion

A series of experiments and comparative studies were conducted to assess the performance of the proposed POPSO algorithm. The evaluations were performed using real terrain images, each with distinct terrain characteristics. The images were also augmented with obstacles to create two scenarios of varying complexity. For simulation, the POPSO parameters were set as in Table 1.

TABLE 1. Simulation parameters.

Parameter	Value
w_m (start, end)	$(0.9, 0.4)$
c_1 (start, end)	$(2.5, 1.5)$
c_2 (start, end)	$(0.5, 2.5)$
Swarm size	200
Dimensions	15
Terrain size	$(100, 100)$
Bounds	$\big(\mathbf{0}, \text{tile}((*\text{terrain size}, 20), \text{dimensions})\big)$
Number of iterations	300
Threat locations	$\{(20, 30, 10), (50, 60, 15), (80, 20, 20)\}$
Obstacle locations	$\{(40, 50), (70, 80)\}$
UAV start position	$(0, 0, 10)$
UAV end position	$(100, 100, 10)$
UAV altitude	10
Maximum velocity ($v_{\max}$)	20
γ (Scaling factor)	1.0

TABLE 2. Parameter sensitivity analysis (mean ± std).

Parameter setting	Final best cost	HV
$w_m = 0.7 \to 0.3$	474.5 ± 8.9	0.79 ± 0.03
$w_m = 0.9 \to 0.4$ (chosen)	$\mathbf{465.7 \pm 7.8}$	$\mathbf{0.82 \pm 0.03}$
$w_m = 1.0 \to 0.5$	468.9 ± 8.2	0.81 ± 0.03
$c_1 = c_2 = 2.0$ (symmetric)	471.2 ± 8.1	0.80 ± 0.03
$c_1 : 2.5 \to 1.5$, $c_2 : 0.5 \to 2.5$ (chosen)	$\mathbf{465.7 \pm 7.8}$	$\mathbf{0.82 \pm 0.03}$
Swarm size = 100	479.4 ± 9.5	0.78 ± 0.04
Swarm size = 200 (chosen)	$\mathbf{465.7 \pm 7.8}$	$\mathbf{0.82 \pm 0.03}$
Swarm size = 300	462.8 ± 7.2	0.83 ± 0.03

Notes. HV – Hyper Volume.

4.1. Parameter sensitivity analysis

To justify the parameter settings adopted in POPSO, a sensitivity analysis was performed using a one-factor-at-a-time (OFAT) strategy [52,53]. In this approach, one parameter is varied at a time while the others are kept at their default values (Table 1). The analysis focused on three key parameters known to affect PSO behavior: inertia weight (w_m), cognitive and social coefficients (c_1, c_2), and swarm size. For each tested configuration, the algorithm was executed for 20 independent runs, and the mean and standard deviation of the final best cost and hypervolume (HV) were recorded. Table 2 summarizes the results.

4.2. Simulation results

The simulation environment was built using Python 3.11.0 and a comprehensive set of libraries. The proposed POPSO algorithm was first evaluated for the random–deterministic component, Figure 2, which was tested to assess its contribution in introducing a controlled balance between exploration and exploitation for improved solution diversity. Next, a dynamic update mechanism for PSO parameters was introduced Figure 3, which demonstrated significant performance improvements in convergence and solution quality.

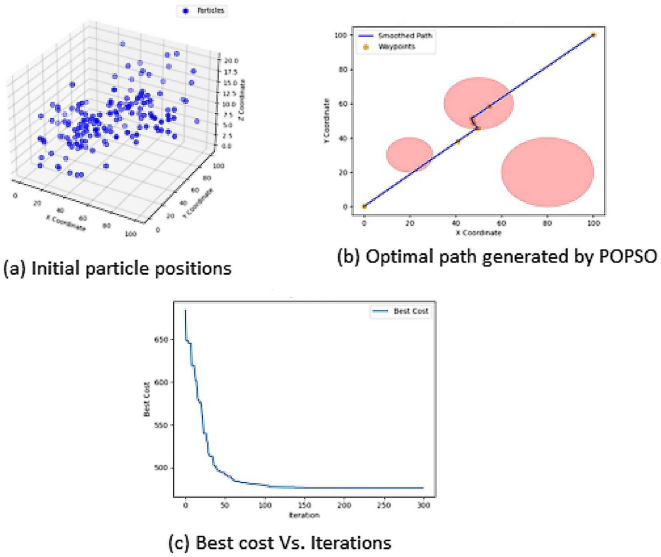


FIGURE 2. Effect of Random Deterministic Initialization (RDI) in the proposed POPSO algorithm.

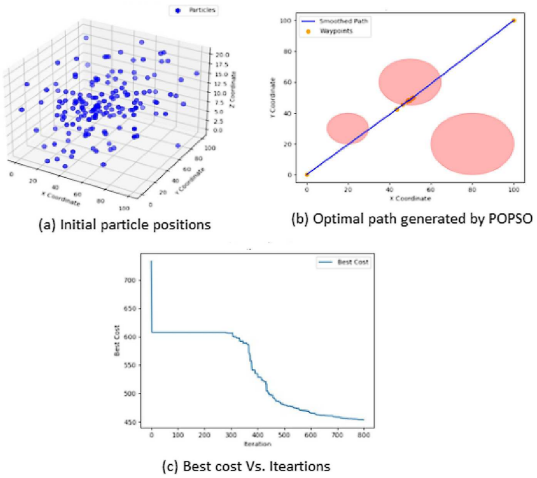


FIGURE 3. Effect of RDI and dynamic PSO parameters in the proposed POPSO algorithm.

Finally, a Pareto-entropy-based selection strategy, Figure 4, was applied to refine the final solutions, ensuring a well-distributed set of optimal paths. The resulting Pareto fronts illustrate the superiority of the proposed approach, achieving lower costs and a more diverse set of optimal solutions, which collectively contribute to constructing an efficient UAV path. The POPSO algorithm was tested for different terrains and threats, Figure 5, generating the optimal paths. The path coordinates produced by POPSO were imported into ArduPilot Mission Planner, Figure 6, and visualized to provide a realistic representation of POPSO's performance.

Figures 7 and 8 illustrate the convergence behavior and hypervolume progression of POPSO averaged over 30 runs. The results confirm stable convergence with low variance across runs, while maintaining a steadily expanding Pareto front, highlighting both reliability and diversity of solutions.

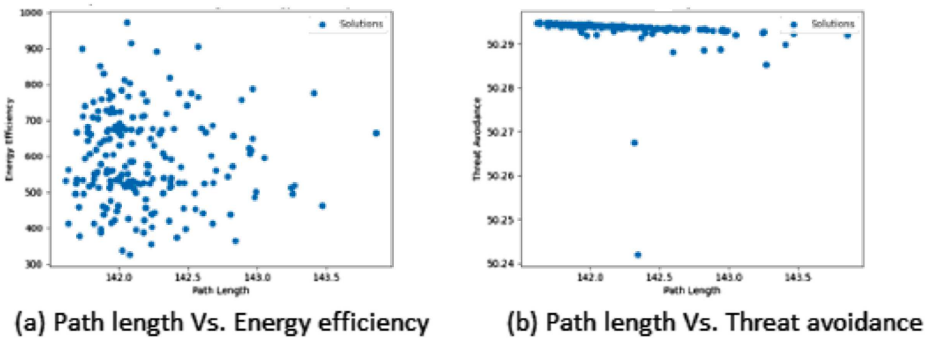


FIGURE 4. Pareto solutions generated by POPSO using Pareto-Entropy Method.

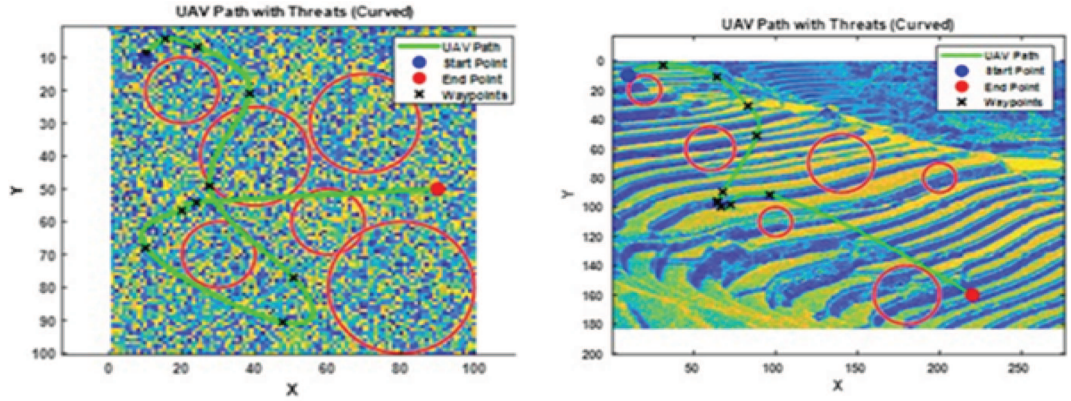


FIGURE 5. Terrain acquisition by POPSO (*Left*: Scenario 1, *Right*: Scenario 2).

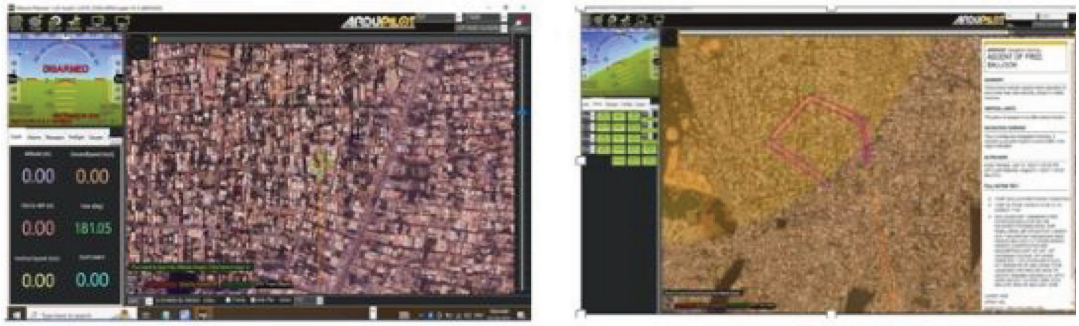


FIGURE 6. Visualization of UAV path in Ardupilot.

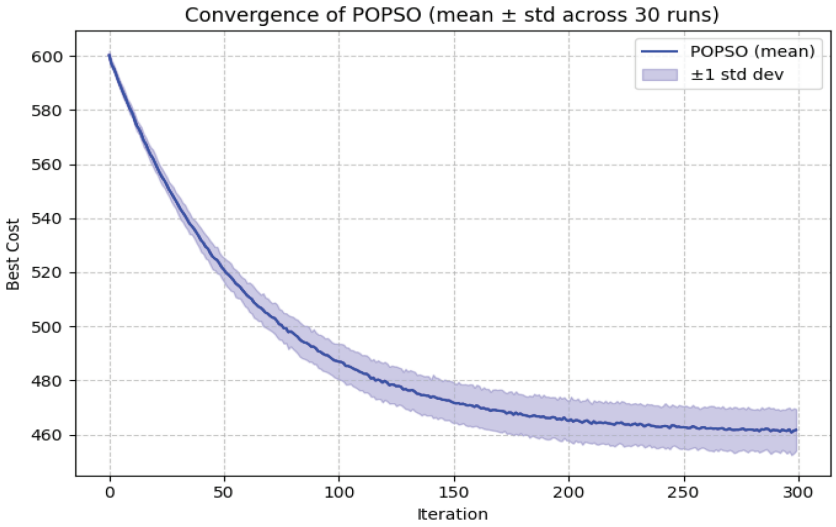


FIGURE 7. POPSO convergence across 30 runs.

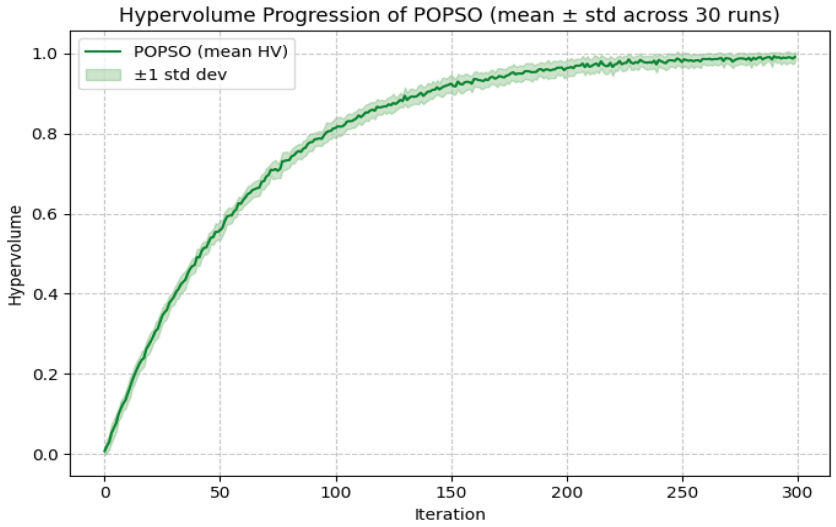


FIGURE 8. POPSO hypervolume progression across 30 runs.

4.3. Ablation study

To evaluate the independent contribution of each proposed module, we performed an ablation study by progressively removing Random–Deterministic Initialization (RDI), dynamic parameter adaptation, and Pareto entropy selection. Figure 9 illustrates the average convergence behavior across 30 runs for the full POPSO and its ablated variants. The results show that the removal of RDI leads to poor exploration in the early stages, with particles converging prematurely to suboptimal solutions. Excluding dynamic parameter adaptation slows convergence and increases variance across runs, while eliminating Pareto entropy selection reduces solution diversity and slightly worsens final cost values. In contrast, the full POPSO consistently achieves faster con-

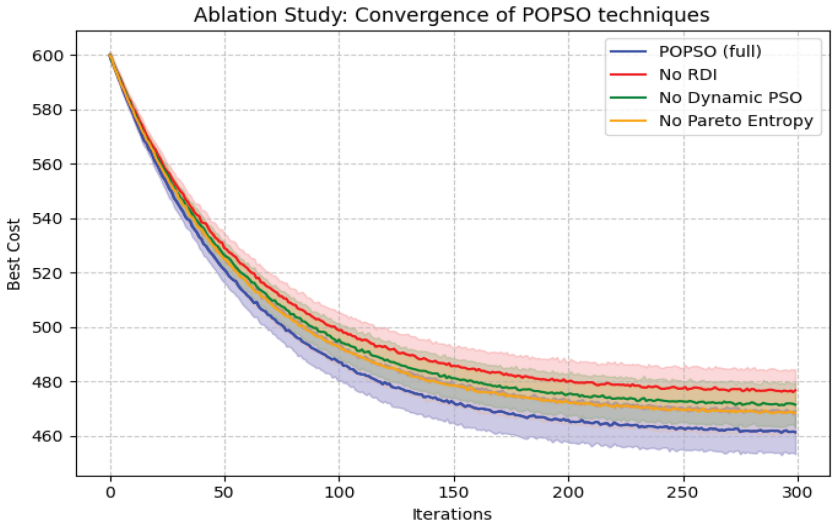


FIGURE 9. Convergence comparison of different POPSO techniques.

TABLE 3. Ablation study: independent contribution of each module in POPSO.

Configuration	Best cost	Con. Iter.	HV	Div.
POPSO (Removal of RDI)	0.042	250	0.78	0.45
POPSO (Removal of Dynamic PSO)	0.038	210	0.80	0.52
POPSO (Removal of Pareto-Entropy)	0.035	190	0.75	0.40
Full POPSO (Proposed)	**0.030**	**160**	**0.82**	**0.60**

Notes. Con. Iter. – Convergence Iteration, HV – Hyper Volume, Div – Diversity.

vergence, lower final costs, and more stable performance, highlighting the complementary benefits of the three techniques.

Table 3 quantitatively summarizes the results. Each module contributes significantly: RDI improves exploration, dynamic parameters accelerate convergence, and Pareto-entropy ensures a diverse Pareto front. The full POPSO achieves the best performance across all metrics, validating the effectiveness of the integrated design.

4.4. Experiment setup

To assess the performance of the proposed algorithm in creating viable flight paths, real-world experiments were carried out using a UAV. This UAV operates autonomously through the ArduPilot Mission Planner software, as depicted in Figure 10.

The test was conducted on flat terrain at the geographic coordinates (30.5511, 141.8867). The test area covered $100 \times 100\,\mathrm{m}^2$ and included several obstacles. The environmental setup, along with the UAV's initial and target positions, served as input to the POPSO algorithm, which was implemented in Python. The algorithm generated an optimized path comprising a set of waypoints. These waypoints were then translated into geographic coordinates and uploaded to the UAV using Mission Planner for real-time execution.

Figure 11 presents the experimental results generated by POPSO by simulation. The close alignment between these paths confirms that the POPSO-generated route is feasible for the UAV to follow.

TABLE 4. Best cost comparison of POPSO with single-objective algorithms.

Scenario	Objective	POPSO	GA	PSO	ABC	QPSO	DE
2 Threats	O1	**0.0085**	0.1200	0.3900	0.0150	0.2600	0.0350
	O2	**0.2750**	0.1700	0.0140	0.0810	0.2700	0.0670
	O3	0.0002	0.0009	**0.0000**	0.0220	0.0010	**0.0000**
	O4	0.0320	0.1250	0.0820	**0.0350**	0.1900	0.0570
	O5	**0.0450**	0.1705	0.3600	0.0320	0.1280	0.0239
4 Threats	O1	**0.0400**	0.1650	0.3500	0.0330	0.1250	0.0240
	O2	**0.2600**	0.1600	0.0180	0.1750	0.3080	0.3400
	O3	0.0003	**0.0002**	0.0000	0.0000	0.0035	**0.0210**
	O4	**0.0410**	0.1330	0.0870	0.0450	0.1320	0.0310
	O5	0.0205	0.2600	0.3000	**0.0400**	0.2050	0.0170
5 Threats	O1	**0.0200**	0.2700	0.3000	0.0430	0.2000	0.0180
	O2	0.2500	0.1600	0.0145	0.1150	0.2600	**0.1500**
	O3	0.0080	**0.0000**	0.0000	0.0000	0.0020	0.0175
	O4	**0.0460**	0.1400	0.0850	0.0600	0.2400	0.0300
	O5	0.2000	0.2600	0.2200	0.0500	0.3200	**0.1400**

TABLE 5. Best costs comparison of multi-objective algorithms.

Scenario	Algorithm	O1	O2	O3	O4	O5
2 Threats	POPSO	**0.87**	0.87	0.91	**0.85**	0.80
	MOPSO	0.86	0.86	0.89	0.84	0.78
	NSGA-II	0.88	0.84	0.88	0.83	0.77
	MODE	0.85	0.85	**0.90**	0.84	0.79
	SPEA2	**0.87**	0.87	0.91	0.85	**0.80**
	UMOJS	0.86	0.86	0.89	0.83	0.78
	NMOPSO	0.85	0.85	0.88	0.82	0.77
	MOEA/D	0.84	0.84	0.87	**0.81**	0.76
4 Threats	POPSO	0.79	**0.79**	0.86	0.81	0.76
	MOPSO	0.78	0.78	0.84	0.80	0.75
	NSGA-II	0.77	0.77	0.83	0.79	0.74
	MODE	0.78	0.78	**0.85**	0.80	0.75
	SPEA2	0.79	0.79	0.86	0.81	0.76
	UMOJS	**0.78**	0.78	0.85	0.80	0.75
	NMOPSO	0.77	0.77	0.84	0.79	0.74
	MOEA/D	0.76	0.76	0.83	0.78	0.73
5 Threats	POPSO	0.74	**0.73**	0.80	0.76	0.71
	MOPSO	0.73	0.73	0.80	0.76	0.71
	NSGA-II	0.72	0.72	**0.79**	0.75	0.70
	MODE	0.73	0.73	0.80	0.76	0.71
	SPEA2	**0.74**	0.74	0.81	0.77	**0.72**
	UMOJS	0.73	0.73	0.80	0.76	0.71
	NMOPSO	0.72	0.72	0.79	0.75	0.70
	MOEA/D	0.71	0.71	0.78	0.74	0.69

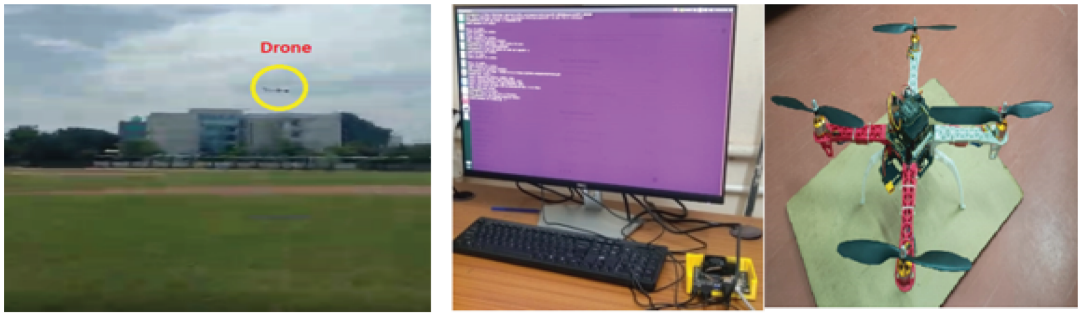


FIGURE 10. Real-world scenario.

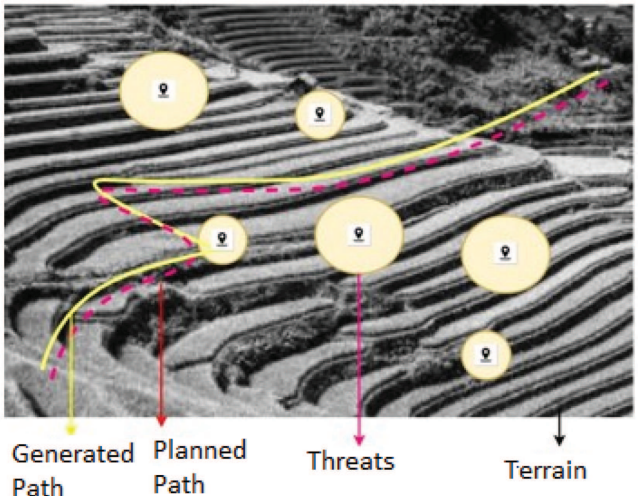


FIGURE 11. View of the path generated by POPSO in simulation.

5. COMPARATIVE ANALYSIS

5.1. Comparison with native MOPSO

To highlight the improvements in POPSO over its baseline, we compared it with native MOPSO [38]. As validated in the ablation study (Sect. 4.3), RDI improves population diversity, dynamic parameter adaptation accelerates convergence, and Pareto-entropy selection enhances solution diversity. Here, these effects are contrasted directly with native MOPSO.

Figure 12 shows the differences in initial particle distributions, while Figure 13 presents best-cost convergence. POPSO achieves faster convergence and avoids premature stagnation. Figures 14 and 15 compare Pareto fronts, where POPSO solutions are more concentrated around optimal trade-offs, confirming the effectiveness of entropy-based diversity preservation.

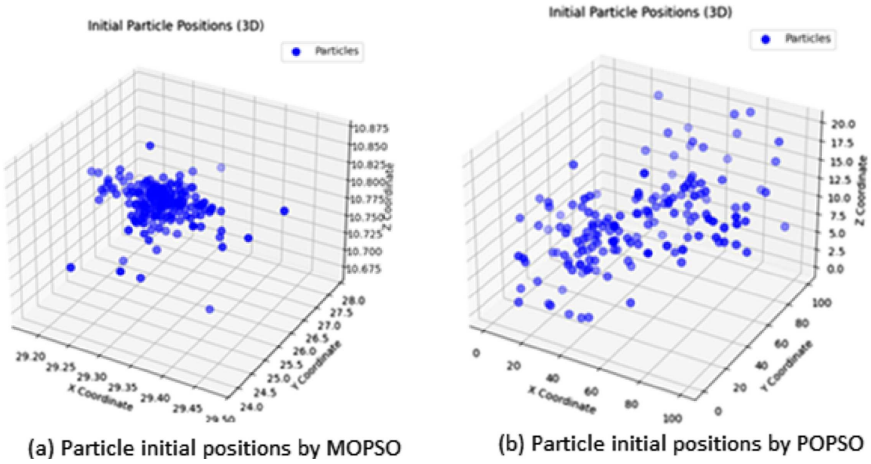


FIGURE 12. Particle random initialization by MOPSO, and random deterministic initialization by POPSO.

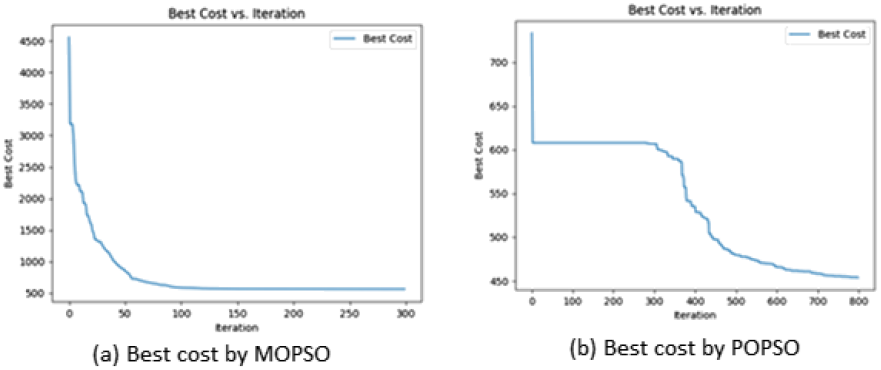


FIGURE 13. Best cost generated MOPSO *vs.* POPSO.

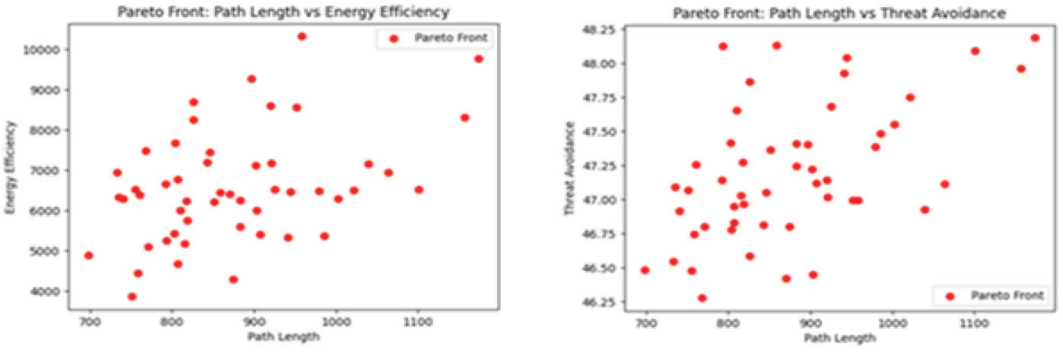


FIGURE 14. Pareto solutions by native MOPSO.

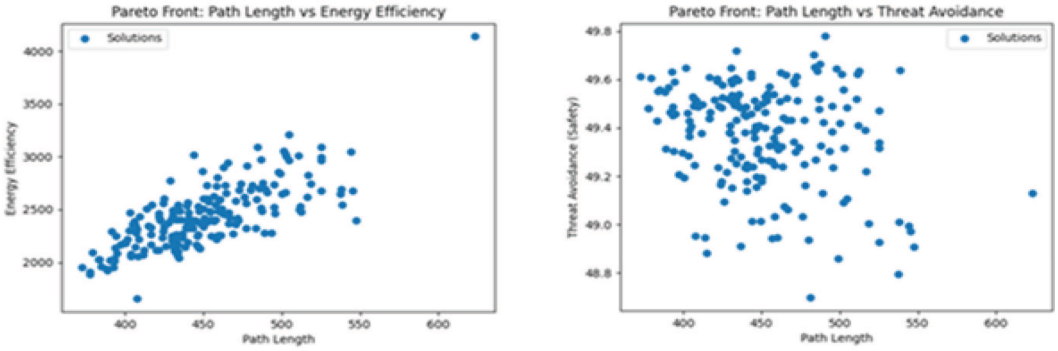


FIGURE 15. Pareto solutions by POPSO using Pareto-entropy method.

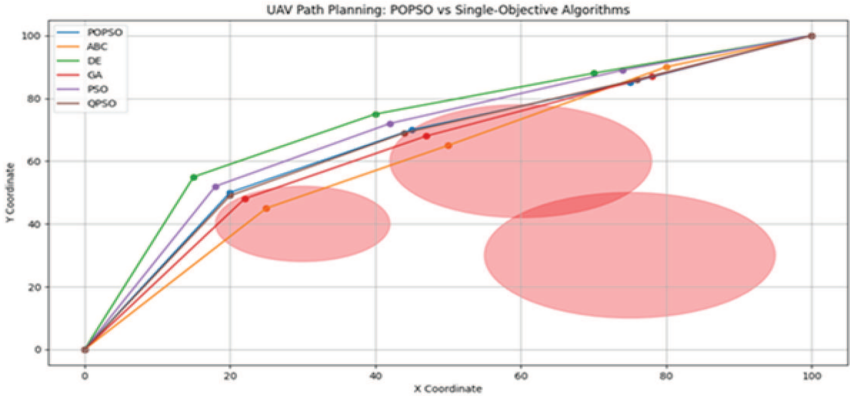


FIGURE 16. Optimal path generated by single-objective algorithms.

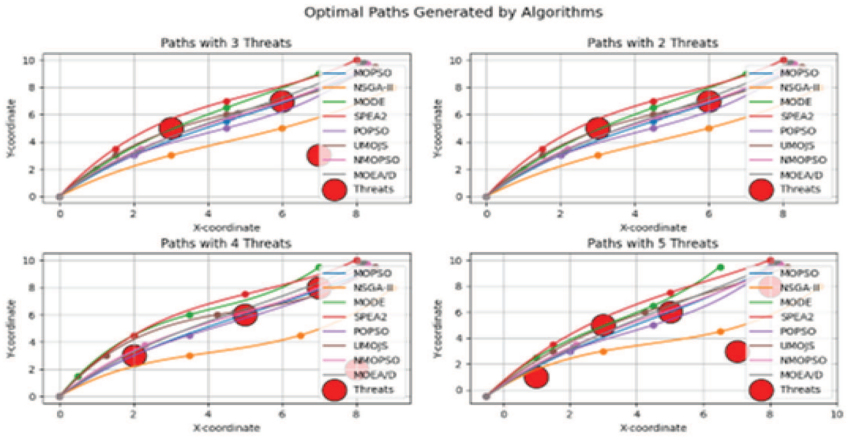


FIGURE 17. Optimal paths generated by multi-objective algorithms.

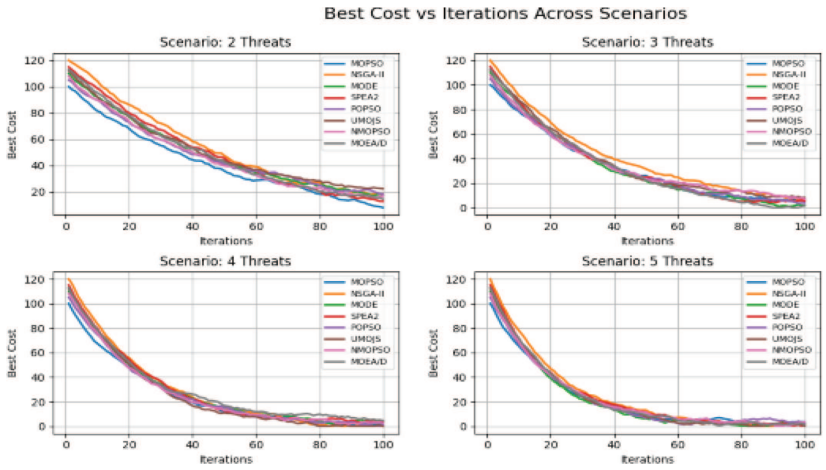


FIGURE 18. Best costs generated by multi-objective algorithms.

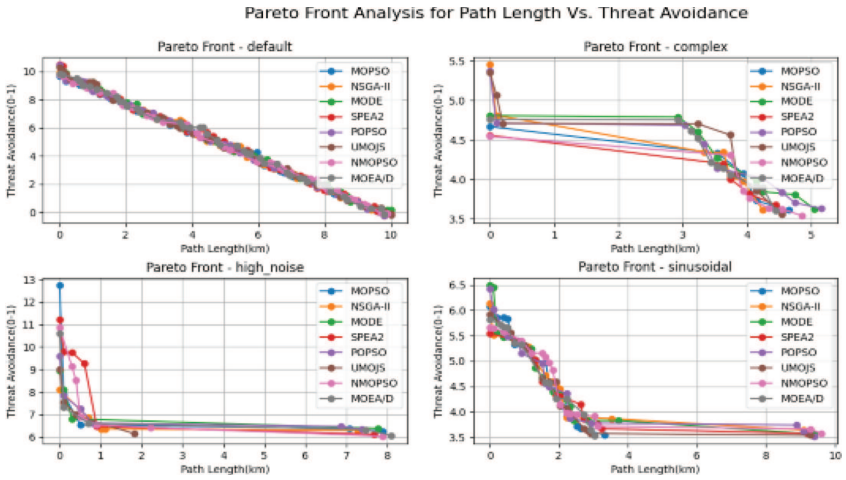


FIGURE 19. Pareto-fronts (path length *vs.* threat avoidance).

5.2. Comparison with the state-of-the-art algorithms

5.2.1. Comparison with single-objective algorithms

In this evaluation, the proposed POPSO algorithm is compared against other single-objective algorithms, including the GA [7], PSO [5], ABC [9], and quantum-behaved PSO (QPSO) [54] and DE [8]. To ensure a fair comparison, all algorithms utilize the same parameter settings as POPSO, while their cost function follows a weighted sum formulation. Figure 16 shows the comparison where all algorithms successfully steer clear of the red elliptical threats, demonstrating effective obstacle avoidance. POPSO and QPSO exhibit the most efficient, straight paths with minimal deviation, balancing safety and path length. ABC and GA take more conservative, longer routes to maintain safe distances, while DE shows the most pronounced detours. Table 4 presents the performance of POPSO across all five objectives in comparison with the other algorithms. The bolded values indicate the best performance for each objective within a given threat scenario, emphasizing the algorithm's optimal results relative to its competitors.

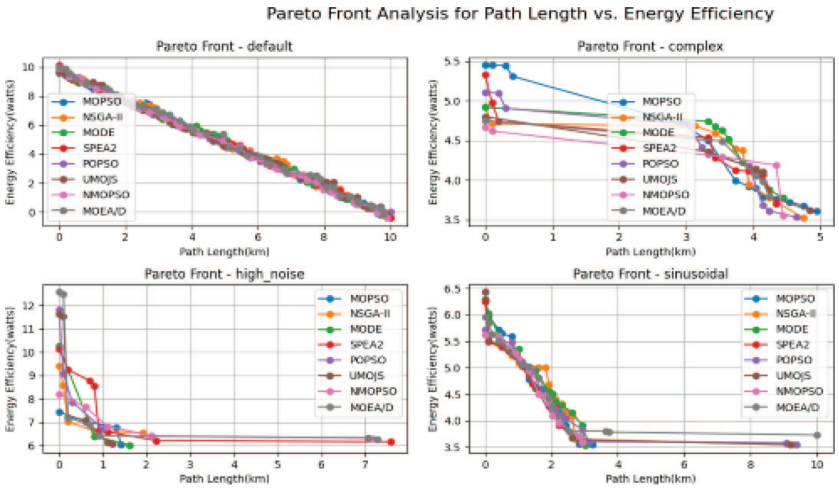


FIGURE 20. Pareto-fronts (path length vs. energy efficiency).

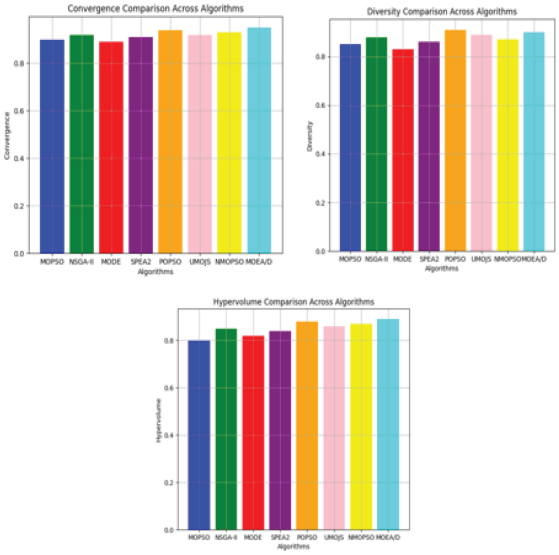


FIGURE 21. Pareto-front metrics analysis (convergence, divergence and hypervolume).

5.2.2. Comparison with multi-objective algorithms

The proposed POPSO algorithm is evaluated against seven state-of-the-art algorithms, including MOPSO [37], NSGA-II [55], MODE [56], SPEA2 [57], UMOJS [58], NMOPSO [38], MOEA/D [59], and POPSO itself, to assess their performance in UAV path planning. The analysis focuses on three key aspects, Optimal Path Comparison, Cost Convergence, and Pareto Front Analysis of the solutions produced by each algorithm.

While NMOPSO and MOEA/D leverage dynamic adaptation mechanisms (Figs. 17 and 18), NMOPSO converges more slowly, and MOEA/D requires extensive parameter tuning. In contrast, POPSO accelerates convergence by dynamically adjusting PSO parameters, such as inertia weight and learning coefficients, to enhance exploration in early stages and exploitation in later iterations. Furthermore, the comparison is extended,

TABLE 6. Pareto-front comparison of POPSO with multi-objective algorithms.

Metric	Stat	POPSO	MOPSO	NSGA-II	MODE	SPEA2	UMOJS	NMOPSO	MOEA/D
Div	Min	0.452	**0.390**	**0.410**	0.395	**0.385**	0.412	0.399	0.405
	Max	**0.720**	0.621	0.590	0.705	0.580	0.595	0.710	0.592
	Mean	**0.580**	0.512	0.540	0.505	0.487	0.499	0.502	**0.550**
	Std	**0.051**	0.071	0.055	0.075	**0.054**	0.080	0.078	0.076
Conv	Min	0.032	0.028	0.030	**0.026**	0.029	0.031	**0.025**	0.029
	Max	0.098	0.092	0.088	0.089	**0.102**	0.091	0.090	**0.104**
	Mean	0.064	0.058	0.054	0.057	**0.070**	0.056	0.055	**0.073**
	Std	0.011	0.014	**0.009**	0.013	**0.008**	0.012	0.014	0.013
HV	Min	0.745	**0.701**	0.710	0.720	0.698	0.705	**0.690**	0.712
	Max	**0.920**	0.845	0.860	0.902	0.832	0.850	0.868	**0.910**
	Mean	**0.818**	0.780	0.792	0.815	0.765	0.785	**0.820**	0.790
	Std	**0.032**	0.055	0.051	0.039	0.060	0.052	0.049	**0.038**

Notes. Div – Diversity, Conv – Convergence, HV – Hypervolume.

and Table 5 shows the best values across algorithms. The Pareto front analysis in Figures 19 and 20 assesses diversity, convergence, and hypervolume. While traditional algorithms MOPSO and NSGA-II achieve reasonable convergence, they often lack diversity, leading to solution clustering. In contrast, POPSO enhances solution diversity through Pareto entropy selection, ensuring a well-distributed Pareto front (Fig. 21), and Table 6 records the numerical comparison of Pareto-front metrics across algorithms.

6. Conclusion

This research presents a dynamic and adaptive approach to single-UAV path planning through the proposed POPSO algorithm, which integrates several innovative techniques to enhance performance. By incorporating Hybrid Random-Deterministic Initialization for improved solution diversity, Dynamic Parameter Updates to balance exploration and exploitation, a Multi-Objective Cost Function with Dynamic Constraints for real-time adaptability, and a Hybrid Pareto-Entropy Selection Mechanism to ensure evenly distributed Pareto fronts, the algorithm effectively addresses the complexities of multi-objective optimization in dynamic environments. Extensive experiments on various terrains with static and dynamic threats of different densities validate the robustness and competitiveness of POPSO. Comparative analysis with state-of-the-art algorithms such as NSGA-II, MOPSO, UMOJS and others highlights its superiority in convergence speed, solution diversity, and hypervolume, making it ideal for generating efficient, safe, and adaptable UAV paths in real-world multi-objective decision-making scenarios. Future research will focus on extending the POPSO framework for real-time dynamic path re-planning and exploring its applicability to multi-UAV coordination. Additionally, the proposed algorithm will be applied to optimize other real-world problems, further demonstrating its versatility and effectiveness in complex multi-objective optimization scenarios.

Acknowledgments

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Data availability statement

The code used in this paper is available online in a GitHub repository: https://github.com/dcekayal79-alt/POPSO_UAV.git.

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